A2 Notes T.4 Radian Measure

$\cos\theta = x$, $\sin\theta = y$, $\tan\theta = \frac{y}{x}$

Warm-Up Use your Unit Circle

1. Find the exact value of $\cos 90°$ in simplest form.

   x

   0

2. Find the exact value of $\sin 150°$ in simplest form.

   $\frac{1}{2}$ y

3. Find the exact value of $\tan 150°$ in simplest form.

   $\frac{\frac{1}{2}}{-\frac{\sqrt{3}}{2}} = \frac{1}{2} \cdot \frac{2}{-\sqrt{3}} = \frac{-1}{\sqrt{3}}\left(\frac{\sqrt{3}}{\sqrt{3}}\right) = \boxed{\frac{-\sqrt{3}}{3}}$

7. Find the exact value of $\tan 30°$ in simplest form.

   $\frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{2} \cdot \frac{2}{\sqrt{3}} = \frac{1}{\sqrt{3}}\left(\frac{\sqrt{3}}{\sqrt{3}}\right) = \boxed{\frac{\sqrt{3}}{3}}$

8. Find the exact value of $\sin 180°$ in simplest form.

   y

   $\boxed{0}$

9. Find the exact value of $\cos 210°$ in simplest form.

   x

   $\boxed{-\frac{\sqrt{3}}{2}}$

Radian Measurement

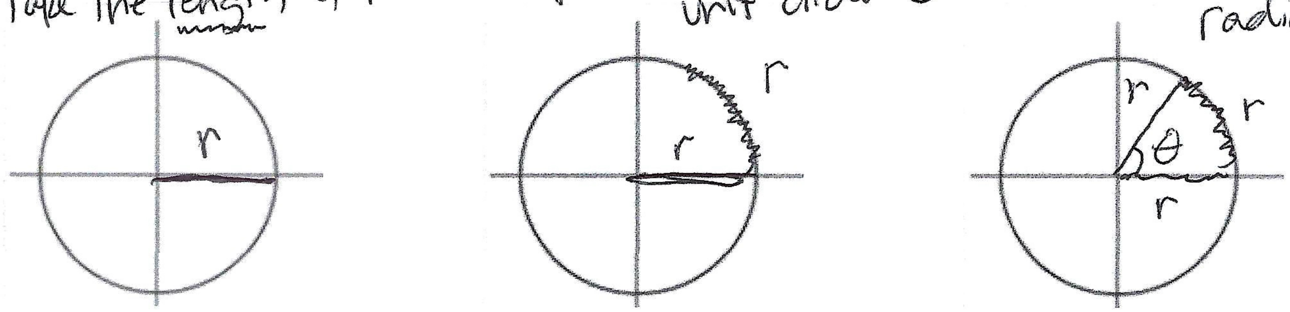
What are radians? Another form of angle measurement (measure of rotation)

Why do we need radians if we already have degrees? it is a unitless form of measurement (no degree symbol)

θ = The angle

Where does it come from?

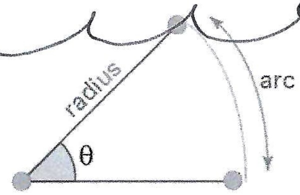
① Take the length of r ② wrap it around the unit circle ③ connect another radius



One radian is the angle that is made at the center of a circle by an arc whose length is equal to the radii

$\theta = 1$ when acr = radius

How are degrees and radians related?



$$\theta = \frac{\text{arc length}}{\text{radius}}$$

The total Circumference of a Circle = $2\pi r$. And the angle representing the entire circle is 360° And $r = 1$ in a unit circle

$\Rightarrow 360^\circ = \frac{2\pi r}{r}$

from $\theta = \frac{\text{arc length (circumference)}}{\text{radius}}$

$\Rightarrow$ Reduce $\Rightarrow$ $360^\circ = 2\pi \Rightarrow$ Divide by 2

$180^\circ = \pi \Rightarrow$ Divide by 180°

$\frac{180^\circ}{180^\circ} = \frac{\pi}{180^\circ}$

$1 = \frac{\pi}{180}$ it's a conversion unit between radians & degrees.

Change 5 feet into inches

$5ft\left(\frac{12in}{1ft}\right)$ ← conversion unit

$= 60$ inches

How to convert between the two types of angle measurement: Use the conversion Unit!

For **degrees to radians**: multiply by $\frac{\pi}{180^\circ}$

For **radians to degrees**: multiply by $\frac{180^\circ}{\pi}$

1. Convert the following angle from degrees to radians. Express your answer in simplest form. Exact

135°

$135^\circ\left(\frac{\pi}{180^\circ}\right) =$ Simplify → 5 goes in to both

$= \frac{27\pi}{36}$ → 9 goes into both $= \boxed{\frac{3\pi}{4}}$

2. Convert the following angle from degrees to radians. Express your answer in simplest form.

180°

$180^\circ\left(\frac{\pi}{180^\circ}\right) = \pi$ "radians"

3. Convert the following angle from degrees to radians. Express your answer in simplest form.

465°

$465^\circ\left(\frac{\pi}{180^\circ}\right) =$ Divide 465 and 180 by 5

$= \frac{93\pi}{36} = \frac{31\pi}{12}$

÷ by 3

6. Convert the angle $\frac{17\pi}{12}$ radians to degrees. so degrees on top

$\frac{17\pi}{12}\left(\frac{180^\circ}{\pi}\right)$ Divide 12 and 180° by 12

$17\cancel{\pi}\left(\frac{15^\circ}{\cancel{\pi}}\right) = 255^\circ$

7. Convert the angle $\frac{7\pi}{6}$ radians to degrees.

$\left(\frac{7\cancel{\pi}}{6}\right)\left(\frac{180^\circ}{\cancel{\pi}}\right)$ 6 goes into 6 and 180°

$7(30^\circ) = 210^\circ$

8. Convert the angle -4 radians to degrees, rounding *to the nearest 10th.*

-4 wait, no π in it?

yes, radians are real #s

$-4\left(\frac{180^\circ}{\pi}\right) = \frac{-720^\circ}{\pi} \approx 229.3^\circ$